

# Causality and instrumental variables

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## Takeaways for this part of class

1. Defining fundamental notions of causal inference:
  - causality
  - structural objects
  - identification
2. Instrumental variables were invented in the context of market equilibrium.
3. The 2SLS estimator can alternatively be written using control-functions or as a Wald ratio.
4. Potential outcomes formalize heterogeneous treatment effects.

# Outline

- Causality and identification.
- Systems of structural equations.
- Alternative representations of the 2SLS estimator.
- Potential outcomes and causality.

# Causal and structural objects in economics

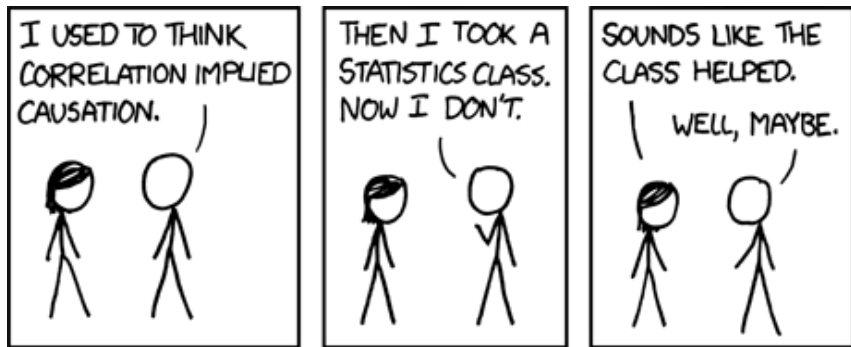
- Returns to schooling
- Elasticity of the tax base with respect to tax rates
- Effect of minimum wage on employment
- Effect of deworming pills on school attendance
- Price elasticity of demand for gasoline
- ...

# Correlation and causality

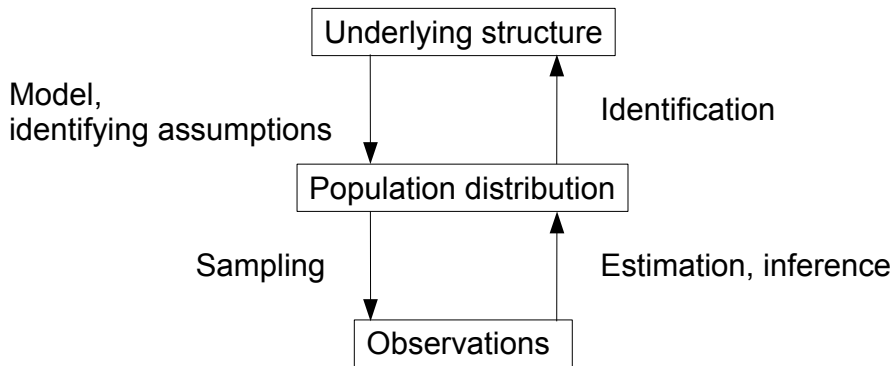
Do observable distributions tell us something about causality?

- College graduates earn  $x\%$  more than high school graduates
- Countries with higher GDP have higher tax rates on average
- Minimum wage levels seem uncorrelated with unemployment levels across time and space
- Gasoline consumption and gasoline price are negatively correlated over time
- ...

Figure: correlation and causation



## Identification vs inference



# Identification vs inference

- goal of econometrics:  
learning interesting things (hopefully)  
about economic phenomena from observations.
- two separate components of econometrics:
  1. identification
  2. estimation and inference



# Estimation and Inference

1. learning about a population distribution
2. from a finite number of observations.
3. examples:
  - estimate a difference in expectations using a difference in means
  - perform inference using a t-test.

# Identification

1. learning about underlying structures, causal mechanisms
  2. from a population distribution.
  3. example:  
    identify a causal effect  
    by a difference in expectations  
    if we have a randomized experiment.
- identification inverts the mapping
  - from underlying structures to a population distribution
  - implied by a model and identifying assumptions.

# Causality

## Practice problem

How would you define causality?

## “Pure” statistics

- causality is meaningless
- observations only tell us about **correlations**
- more generally, joint distributions
- disclaimer: few statisticians today would say this!

# Sciences

- Galileo Galilei: one of the first to follow experimental ideal
- full **control of experimental circumstances**.
- do the same thing  
⇒ same thing happens to the outcomes you measure
- variation in experimental circumstances  
⇒ difference in observed outcomes  $\approx$  causal effect
- example:
  - dropping a ball from different floors of the tower of Pisa
  - different time till the ball hits the ground
- crucial component:  
**external intervention** ( “exogenous variation” )  
⇒ allows to interpret correlation as causation

# Social and biological sciences

- economics is not physics
- this version of the experimental ideal is not very useful
- reason: many unobserved, and unknown, factors which we cannot hope to control
- $\Rightarrow$  **can never replicate** experimental circumstances fully
- there is “**unobserved heterogeneity.**”

## Social and biological sciences ctd

- not all is lost, however
- can still hope create experimental circumstances which are the same **on average**
- this is the idea of a **randomized experiment!**
- randomly pick treatment and control groups  
⇒ they are identical on average.
- “compare apples with apples.”
- many settings of interest in economics:  
not possible to run experiments
- but: definition of causality is intimately tied to randomized experiment,  
hypothetical or actual

# Recap

- Framework of classic probability theory:
  - Does not allow to talk about causality,
  - only joint distributions.
- Causality in the sciences:
  - Additional concept:
  - External intervention / exogeneous variation
  - $\Rightarrow$  experiments.
- Causality in econometrics, biostatistics,...:
  - Additional concept:
  - Unobserved heterogeneity.
  - $\Rightarrow$  randomized experiments
  - (or “quasi-experiments”)



# Structural objects

## Practice problem

How would you define “structural?”

## My preferred definition

- An object is structural, if it is **invariant** across relevant counterfactuals.
- Example: dropping a ball from tower of Pisa
  - acceleration is the same, no matter which floor you drop it from
  - also the same if you do this on the Eiffel tower
  - time to ground would not be the same
  - acceleration is not the same if you do this on the moon

# Possibly structural objects

- “economic primitives”
  - preferences
  - technologies
  - assumed to be invariant across policy changes
- derived objects
  - demand function, as supply, and thus price, varies
  - more generally causal effects, as treatment varies

## Another common use for “structural”

- full specification of parametric forms for economic primitives
- assumptions strong enough to identify these primitives
- somewhat misleading terminology:
  1. less ambitious approaches with weaker assumptions can also identify structural objects
  2. if assumptions for full identification of economic primitives are violated then the objects identified are not structural.

## Origins: systems of structural equations

- econometrics pioneered by “Cowles commission” starting in the 1930s
- they were interested in demand (elasticities) for agricultural goods
- introduced systems of simultaneous equations
  - outcomes as equilibria of some structural relationships
  - goal: recover the slopes of structural relationships
  - from observations of equilibrium outcomes and exogenous shifters

## System of structural equations

$$Y = A \cdot Y + B \cdot Z + U,$$

- $Y$ :  $k$ -dimensional vector of equilibrium outcomes
- $Z$ :  $l$ -dimensional vector of exogenous variables
- $A$ : unknown  $k \times k$  matrix of coefficients of interest
- $B$ : unknown  $k \times l$  matrix
- $U$ : further unobserved factors affecting outcomes

## Example: supply and demand

$$Y = (P, Q)$$

$$P = A_{12} \cdot Q + B_1 \cdot Z + U_1 \text{ demand}$$

$$Q = A_{21} \cdot P + B_2 \cdot Z + U_2 \text{ supply}$$

- demand function: relates prices to quantity supplied and shifters  $Z$  and  $U_1$  of demand
- supply function relates quantities supplied to prices and shifters  $Z$  and  $U_2$  of supply.
- does not really matter which of the equations puts prices on the “left hand side.”
- price and quantity in market equilibrium: solution of this system of equations.

## Reduced form

- solve equation  $Y = A \cdot Y + B \cdot Z + U$   
for  $Y$  as a function of  $Z$  and  $U$
- bring  $A \cdot Y$  to the left hand side,  
pre-multiply by  $(I - A)^{-1} \Rightarrow$

$$Y = C \cdot Z + \eta \text{ “reduced form”}$$

$$C := (I - A)^{-1} \cdot B \text{ reduced form coefficients}$$

$$\eta := (I - A)^{-1} \cdot U$$

- suppose  $E[U|Z] = 0$  (ie.,  $Z$  is randomly assigned)
- then we can **identify**  $C$  from

$$E[Y|Z] = C \cdot Z.$$



## Exclusion restrictions

- suppose we know  $C$
- what we want is  $A$ , possibly  $B$
- problem:  $k \times l$  coefficients in  $C = (I - A)^{-1} \cdot B$   
 $k \times (k + l)$  coefficients in  $A$  and  $B$
- $\Rightarrow$  further assumptions needed
- exclusion restrictions: assume that some of the coefficients in  $B$  or  $A$  are  $= 0$ .
- Example: rainfall affects grain supply but not grain demand

## Supply and demand continued

- suppose  $Z$  is (i) random,  $E[U|Z] = 0$
- and (ii) “excluded” from the demand equation  
 $\Rightarrow B_{11} = 0$
- by construction,  $\text{diag}(A) = 0$
- therefore

$$\text{Cov}(Z, P) = \text{Cov}(Z, A_{12} \cdot Q + B_1 \cdot Z + U_1) = A_{12} \cdot \text{Cov}(Z, Q),$$

- $\Rightarrow$  the slope of demand is identified by

$$A_{12} = \frac{\text{Cov}(Z, P)}{\text{Cov}(Z, Q)}.$$

- $Z$  is an **instrumental variable**

## Alternative ways of writing the linear IV estimand

- Linear triangular system:  $Y, D \in \mathbb{R}$ ,

$$Y = \beta_0 + \beta_1 D + U$$

$$D = \gamma_0 + \gamma_1 Z + V.$$

- Relation to system of structural equations:
  - $Y$  equation: structural equation (say first line in system)
  - $D$  equation: reduced form relationship
- Exogeneity conditions (subsume exclusion, randomization), relevance condition:

$$\text{Cov}(Z, U) = 0$$

$$\text{Cov}(Z, V) = 0$$

$$\text{Cov}(Z, D) = \gamma_1 \text{Var}(Z) \neq 0.$$

- Under these conditions,

$$\beta_1 = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)}.$$

## Two-stage least squares

- Let  $\hat{D}$  be the predicted value from a first stage regression,

$$\hat{D} = \gamma_0 + \gamma_1 Z,$$

- and regress  $Y = \beta_0 + \beta_1 D + U$  on  $\hat{D}$ ,

$$Y = \alpha_0 + \alpha_1 \hat{D} + \tilde{U}.$$

- Then, since  $\text{Cov}(Z, U) = 0$ ,

$$\alpha_1 = \frac{\text{Cov}(\hat{D}, Y)}{\text{Var}(\hat{D})} = \frac{\gamma_1 \text{Cov}(Z, Y)}{\gamma_1^2 \text{Var}(Z)} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \beta_1.$$

- $\Rightarrow$  two-stage least squares identifies  $\beta_1$ !

## Control function

- $V$  is the residual of a first stage regression of  $D$  on  $Z$ .
- Consider a regression of  $Y$  on  $D$  and  $V$ ,

$$Y = \delta_0 + \delta_1 D + \delta_2 V + W$$

- Partial regression formula:
  - $\delta_1$  is the coefficient of a regression of  $\tilde{Y}$  on  $\tilde{D}$  (or of  $Y$  on  $\tilde{D}$ ),
  - where  $\tilde{Y}$ ,  $\tilde{D}$  are the residuals of regressions on  $V$ .
- By construction:

$$\tilde{D} = \gamma_0 + \gamma_1 Z = \hat{D}$$

$$\tilde{Y} = \beta_0 + \beta_1 \tilde{D} + \tilde{U}$$

- $\text{Cov}(Z, U) = \text{Cov}(Z, V) = 0$  implies  $\text{Cov}(\tilde{D}, \tilde{U}) = 0$ , and thus

$$\delta_1 = \beta_1.$$

# Recap

- Three numerically equivalent estimands:

1. The slope

$$\text{Cov}(Z, Y) / \text{Cov}(Z, D).$$

2. The two-stage least squares slope from the regression

$$Y = \alpha_0 + \alpha_1 \hat{D} + \tilde{U},$$

where  $\tilde{U} = (\beta_1 V + U)$ , and  $\hat{D}$  is the first stage predicted value  $\hat{D} = \gamma_0 + \gamma_1 Z$ .

3. The slope of the regression with control

$$Y = \delta_0 + \delta_1 D + \delta_2 V + W,$$

where the control function  $V$  is given by the first stage residual,  $V = D - \gamma_0 - \gamma_1 Z$ .

## Empirical example

- **Aizer, A. and Doyle, J. J. (2015).** Juvenile incarceration, human capital, and future crime: Evidence from randomly assigned judges. *The Quarterly Journal of Economics*, 130(2):759–803.
- Judges (within days, courts) randomly assigned to cases.
- Use judge-specific incarceration rate as instrument for juvenile incarceration.
- Finding: Juvenile incarceration reduces high school graduation, increases adult crime.

TABLE III  
FIRST STAGE

|                                                                      | (1)              | (2)              | (3)              |
|----------------------------------------------------------------------|------------------|------------------|------------------|
| Dependent variable: juvenile<br>incarcerations                       |                  | OLS              |                  |
| First judge's leave-out mean incarceration<br>rate among first cases | 1.103<br>(0.102) | 1.082<br>(0.095) | 1.060<br>(0.097) |
| Demographic controls                                                 | No               | Yes              | Yes              |
| Court controls                                                       | No               | No               | Yes              |
| Observations                                                         | 37,692           |                  |                  |
| Mean of dependent variable                                           | 0.227            |                  |                  |

*Notes.* This table reports the first-stage relationship between juvenile incarceration and the instrument: the judge's incarceration rate using the linked Chicago Public School–Juvenile Court of Cook County data including cases from 1990–2000 as described in the text. All models include community  $\times$  weapons-offense  $\times$  year-of-offense fixed effects. Demographic controls include indicators for four



TABLE IV  
JUVENILE INCARCERATION AND HIGH SCHOOL GRADUATION

|                            | Dependent variable: graduated high school |                    |                                       |                       |                    |                   |                   |
|----------------------------|-------------------------------------------|--------------------|---------------------------------------|-----------------------|--------------------|-------------------|-------------------|
|                            | (1)                                       | (2)                | (3)                                   | (4)                   | (5)                | (6)               | (7)               |
|                            | Full CPS sample                           |                    |                                       | Juvenile court sample |                    |                   |                   |
|                            | OLS                                       | OLS                | Inverse propensity<br>score weighting | OLS                   | OLS                | 2SLS              | 2SLS              |
| Juvenile incarceration     | -0.389<br>(0.0066)                        | -0.292<br>(0.0065) | -0.391<br>(0.0055)                    | -0.088<br>(0.0043)    | -0.073<br>(0.0041) | -0.108<br>(0.044) | -0.125<br>(0.043) |
| Demographic controls       | No                                        | Yes                | Yes                                   | No                    | Yes                | No                | Yes               |
| Court controls             | N/A                                       | N/A                | N/A                                   | No                    | Yes                | No                | Yes               |
| Observations               | 440,797                                   | 440,797            | 420,033                               | 37,692                |                    |                   |                   |
| Mean of dependent variable | 0.428                                     | 0.428              | 0.433                                 | 0.099                 |                    |                   |                   |

*Notes.* This table reports the relationship between juvenile incarceration and graduation from Chicago Public Schools. Columns (1)–(3) include all students in Chicago Public Schools in eighth grade during 1990–2006 and at least age 25 by 2008. Columns (1) and (2) include community fixed effects, while column (2) also includes indicators for race, sex, special education status, each year of birth, and the 2000 U.S. census tract family poverty rate. Column (3) used the same controls and community indicators to calculate the propensity score using a probit model, estimated on a subsample where probit estimation is possible (where there is variation in juvenile incarceration within cells). Columns (4)–(7) use the linked Chicago Public School–Juvenile Court of Cook County data including cases from 1990–2000 as described in the text. These models include community  $\times$  weapons-offense  $\times$  year-of-offense fixed effects. Demographic controls include those listed for column (2). Court controls include nine offense categories, indicators for seven risk-assessment index categories, and whether the first judge assigned was missing. Standard errors are reported in the parentheses and are clustered at the community level. The propensity score standard errors were calculated using 200 bootstrap replications.

TABLE V  
JUVENILE INCARCERATION AND ADULT CRIME

|                            | Dependent variable: entered adult prison by age 25 |                   |                                       |                       |                   |                  |                  |
|----------------------------|----------------------------------------------------|-------------------|---------------------------------------|-----------------------|-------------------|------------------|------------------|
|                            | (1)                                                | (2)               | (3)                                   | (4)                   | (5)               | (6)              | (7)              |
|                            | Full CPS sample                                    |                   |                                       | Juvenile court sample |                   |                  |                  |
|                            | OLS                                                | OLS               | Inverse propensity<br>score weighting | OLS                   | OLS               | 2SLS             | 2SLS             |
| Juvenile incarceration     | 0.407<br>(0.0082)                                  | 0.350<br>(0.0064) | 0.219<br>(0.013)                      | 0.200<br>(0.0072)     | 0.155<br>(0.0073) | 0.260<br>(0.073) | 0.234<br>(0.076) |
| Demographic controls       | No                                                 | Yes               | Yes                                   | No                    | Yes               | No               | Yes              |
| Court controls             | N/A                                                | N/A               | N/A                                   | No                    | Yes               | No               | Yes              |
| Observations               | 440797                                             | 440797            | 420033                                | 37692                 |                   |                  |                  |
| Mean of dependent variable | 0.067                                              | 0.067             | 0.057                                 | 0.327                 |                   |                  |                  |

*Notes.* This table reports the relationship between juvenile incarceration and imprisonment in an adult facility by the age of 25. Columns (1)–(3) include all students in Chicago Public Schools in eighth grade during 1990–2006 and at least age 25 by 2008. Columns (1) and (2) include community fixed effects, while column (2) also includes indicators for race, sex, special education status, each year of birth, and the 2000 U.S. census tract family poverty rate. Column (3) used the same controls and community indicators to calculate the propensity score using a probit model, estimated on a subsample where probit estimation is possible (where there is variation in juvenile incarceration within cells). Columns (4)–(7) use the linked Chicago Public School–Juvenile Court of Cook County–Illinois Department of Corrections data including juvenile cases from 1990–2000 as described in the text. These models include community  $\times$  weapons-offense  $\times$  year-of-offense fixed effects. Demographic controls include those listed for column (2). Court controls include nine offense categories, indicators for seven risk-assessment index categories, and whether the first judge assigned was missing. Standard errors are reported in the parentheses and are clustered at the community level. The propensity score standard errors were calculated using 200 bootstrap replications.

## Remarks

- historically, applied researchers have not been very careful about choosing  $Z$  for which
  - (i) randomization and (ii) exclusion restriction are well justified.
- since the 1980s, more emphasis on credibility of identifying assumptions
- some additional problematic restrictions we imposed:
  1. linearity
  2. constant (non-random) slopes
  3. heterogeneity is  $k$  dimensional and enters additively
- $\Rightarrow$  causal effects assumed to be the same for everyone
- next: framework which does not impose this

## Treatment effects and potential outcomes

- coming from biostatistics / medical trials
- potential outcome framework: answer to “what if” questions
- two “treatments:”  $D = 0$  or  $D = 1$
- eg. placebo vs. actual treatment in a medical trial
- $Y_i$  person  $i$ 's outcome  
eg. survival after 2 years
- potential outcome  $Y_i^0$ :  
what if person  $i$  would have gotten treatment 0
- potential outcome  $Y_i^1$ :  
what if person  $i$  would have gotten treatment 1
- question to you: is this even meaningful?

- causal effect / treatment effect for person  $i$  :  
 $Y_i^1 - Y_i^0$ .

- average causal effect / average treatment effect:

$$ATE = E[Y^1 - Y^0],$$

- expectation averages over the population of interest

# The fundamental problem of causal inference

- **we never observe both  $Y^0$  and  $Y^1$  at the same time**
- one of the potential outcomes is always missing from the data
- treatment  $D$  determines which of the two we observe
- formally:

$$Y = D \cdot Y^1 + (1 - D) \cdot Y^0.$$

## Selection problem

- distribution of  $Y^1$  among those with  $D = 1$   
need not be the same as the distribution of  $Y^1$  among everyone.
- in particular

$$E[Y|D = 1] = E[Y^1|D = 1] \neq E[Y^1]$$

$$E[Y|D = 0] = E[Y^0|D = 0] \neq E[Y^0]$$

$$E[Y|D = 1] - E[Y|D = 0] \neq E[Y^1 - Y^0] = ATE.$$

# Randomization

- no selection  $\Leftrightarrow D$  is random

$$(Y^0, Y^1) \perp D.$$

- in this case,

$$E[Y|D=1] = E[Y^1|D=1] = E[Y^1]$$

$$E[Y|D=0] = E[Y^0|D=0] = E[Y^0]$$

$$E[Y|D=1] - E[Y|D=0] = E[Y^1 - Y^0] = ATE.$$

- can ensure this by actually randomly assigning  $D$
- independence  $\Rightarrow$  comparing treatment and control actually compares “apples with apples”
- this gives **empirical content to** the “metaphysical” notion of **potential outcomes!**



## Empirical example

- **Bertrand, M. and Mullainathan, S. (2004).** Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination. *American Economic Review*, 94(4):991–1013.
- Randomly assign names which are statistically “white” or “black” to resumes which are sent out as job applications.
- Estimate causal effect on likelihood of getting invited to a job interview.

TABLE 4—AVERAGE CALLBACK RATES BY RACIAL SOUNDINGNESS OF NAMES AND RESUME QUALITY

| Panel A: Subjective Measure of Quality<br>(Percent Callback) |                 |                  |       |                               |
|--------------------------------------------------------------|-----------------|------------------|-------|-------------------------------|
|                                                              | Low             | High             | Ratio | Difference ( <i>p</i> -value) |
| White names                                                  | 8.50<br>[1,212] | 10.79<br>[1,223] | 1.27  | 2.29<br>(0.0557)              |
| African-American names                                       | 6.19<br>[1,212] | 6.70<br>[1,223]  | 1.08  | 0.51<br>(0.6084)              |
| Panel B: Predicted Measure of Quality<br>(Percent Callback)  |                 |                  |       |                               |
|                                                              | Low             | High             | Ratio | Difference ( <i>p</i> -value) |
| White names                                                  | 7.18<br>[822]   | 13.60<br>[816]   | 1.89  | 6.42<br>(0.0000)              |
| African-American names                                       | 5.37<br>[819]   | 8.60<br>[814]    | 1.60  | 3.23<br>(0.0104)              |

*Notes:* Panel A reports the mean callback percents for applicant with a White name (row 1) and African-American name (row 2) depending on whether the resume was subjectively qualified as a lower quality or higher quality. In brackets is the number of resumes sent for each race/quality group. The last column reports the *p*-value of a test of proportion testing the null hypothesis that the callback rates are equal across quality groups within each racial group. For Panel B, we use a third of the sample to estimate a probit regression of the callback dummy on the set of resume characteristics as displayed in Table 3. We further control for a sex dummy, a city dummy, six occupation dummies, and a vector of dummy variables for job requirements as listed in the employment ad (see Section III, subsection D, for details). We then use the estimated coefficients on the set of resume characteristics to estimate a predicted callback for the remaining resumes (two-thirds of the sample). We call “high-quality” resumes the resumes that rank above the median predicted callback and “low-quality” resumes the resumes that rank below the median predicted callback. In

## Relation to linear structural equations?

- Linear structural equations are a special case of treatment effect framework.
- Suppose

$$Y = \alpha + \beta D + U,$$

- where this equation is “structural.”  
 $\Leftrightarrow$  Coefficients and the residual stay the same if treatment is changed.
- Then

$$Y^0 = \alpha + U$$

$$Y^1 = \alpha + \beta + U,$$

- and  $ATE = \beta$ .
- Note: Causal effects are the same for everyone in the linear framework.

# Nonlinear structural equations

- without imposing linearity / restrictions on the dimension of heterogeneity:

$$Y = g(D, U).$$

- equivalent to potential outcomes
- potential outcomes notation: more compact
- structural function notation: more explicit  
which determinants are held constant to define causal effects in more complex settings? Eg.

$$Y = g(D, X, W, U)$$